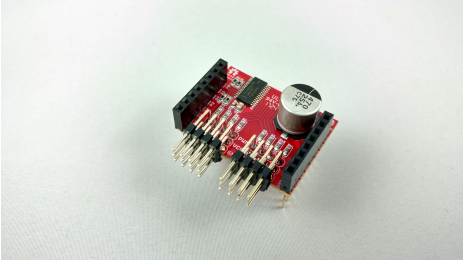
OpenMV® Servo Shield (V1)

Datasheet

SKU: OMV-SHIELD-SERVO-V1

Description



The Servo Shield V1 enables the OpenMV Cam to control up to 8 servos simultaneously using the PCA9685 servo controller, which also supports general-purpose PWM functionality. Two example scripts are provided for controlling servos and general-purpose PWM. This is the retired V1 version — see the current Servo Shield V2/V3 for the latest design.

Target Areas

Robotics, servo control, PWM generation, multi-servo systems, pan-tilt mechanisms.

Key Features

- **8-channel servo control** — Control up to 8 servos simultaneously via PCA9685 servo controller.
- **General-purpose PWM** — PCA9685 also supports general-purpose PWM output.
- **Example scripts** — PCA9685.py for PWM control and Servo.py for servo control included.
- **Shield stacking** — Compatible with Motor Shield and Pan and Tilt Shield (different pins).

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1 Features

1.1 Servo Controller

Component	Details
Chip	PCA9685 — 16-channel, 12-bit PWM controller via I2C.
Servo Channels	Up to 8 servos simultaneously.
PWM	Also supports general-purpose PWM output.

1.2 Compatibility

Component	Details
Supported Cameras	All OpenMV Cam models.
Shield Stacking	Compatible with Motor Shield and Pan and Tilt Shield (different pins).

2 General Info

Dimension	Value
Weight	30 g

3 Company Information

Field	Details
Website	https://openmv.io

4 Reference Documentation

Reference	Link / Document
Product Page	https://openmv.io/products/servo-shield
PCA9685.py Example	https://github.com/openmv/openmv/blob/master/scripts/examples/50-OpenMV-Boards/60-Shields/61-Servo-Shield/pca9685.py
Servo.py Example	https://github.com/openmv/openmv/blob/master/scripts/examples/50-OpenMV-Boards/60-Shields/61-Servo-Shield/servo.py
Documentation	https://docs.openmv.io

5 Revision History

Date	Revision	Changes
2026-04-04	Rev 1	Initial public release.